BeSA:
Benchmarks for Simulating Agent-based Models in Architecture Research

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November 1, 2026

The Fourth Workshop on Architecture for Health, MICRO 2026

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https://s4lresearch.org

Executive Summary

- Agent-based modeling (ABM) is a powerful simulation tool used in many fields including health

- **Problem:**
  - ABMs do not work efficiently and do not scale well
  - No prior work identifies ABM's performance bottlenecks
  - No benchmark suite exists for architecture research on accelerating and scaling ABMs

- **Goal:** Understand and improve the performance of large-scale agent-based simulations

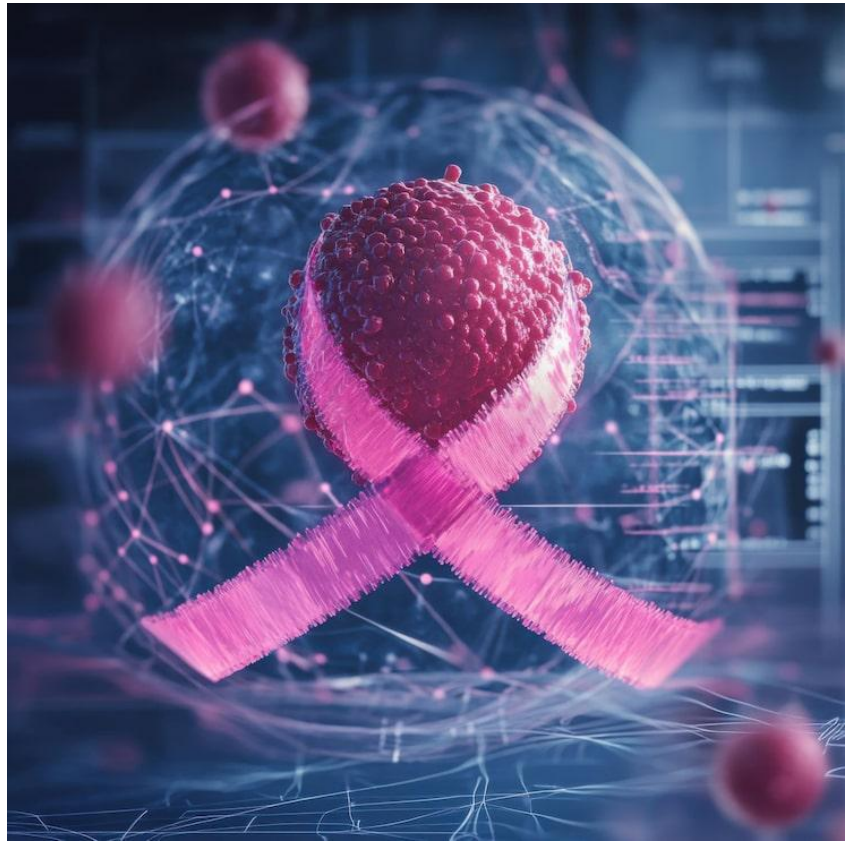
- **Key Observations:**
  - Short computation wrapped around irregular walks over a neighbor structure
  - Most of the execution time goes to waiting for memory
  - Limited by the memory capacity

- **BeSA:**
  - The first ABM benchmark suite for architecture research (work in progress)
  - Covers different health use cases
  - Compatible with gem5 simulator

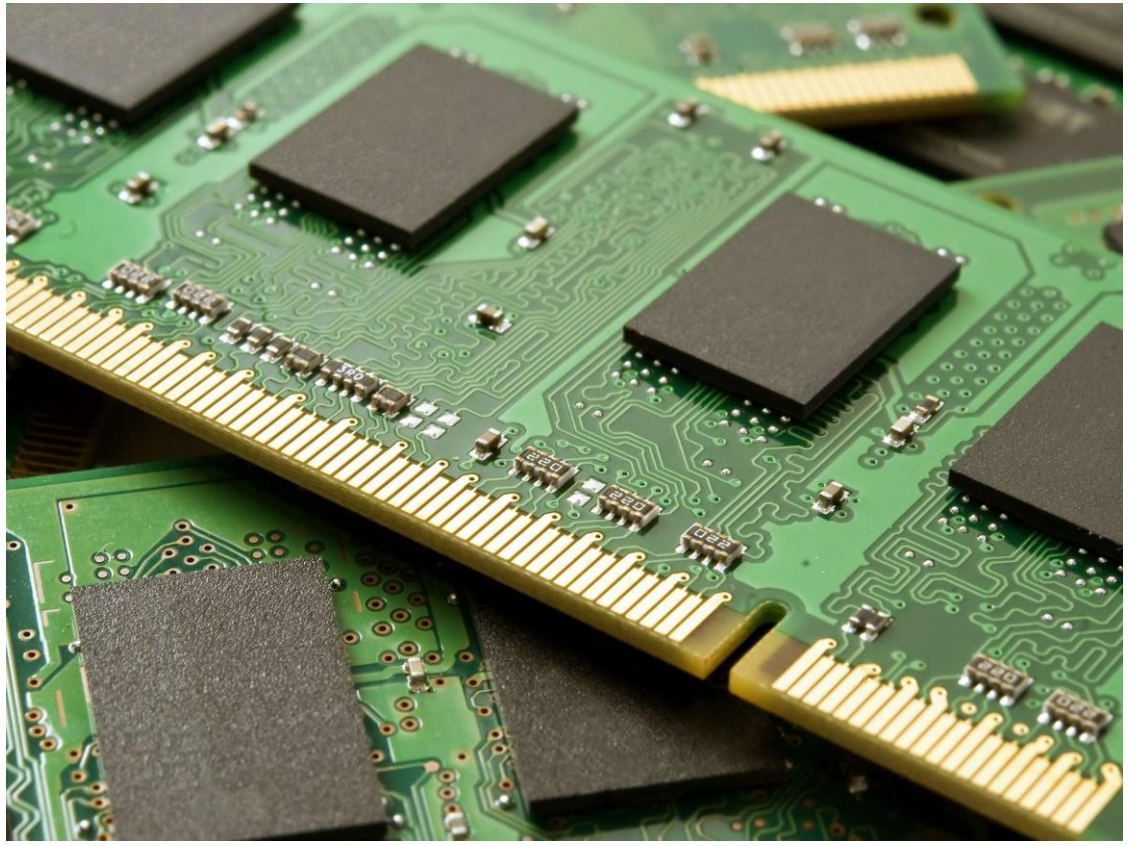
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Two Different Types of Research

Cancer Research



Computer Architecture Research



Two Different Types of Research

Cancer Research

- Very complex biological systems
- Experiments take months or years
- Very expensive
- Low statistical significance
  - Reproducing results is expensive
  - Small sample sets
  - Inevitable variations
  - Low fidelity of metrics

Slow and expensive scientific progress

Computer Architecture Research

- Less complex
- Just hours
- Not expensive
- Higher statistical significance
  - Reproducibility is low-cost
  - Larger sample sets
  - Controllable variations
  - High fidelity of metrics

Fast and low-cost scientific progress

Simulations enable
fast and low-cost scientific progress
in computer architecture research

Why not leverage them for cancer research?

Bad news:

Accelerating scientific progress demands

low-cost and fast simulations of ***complex systems***

Ugly news:

Cancer research is just one example;
there are many others

Good news:

Yes, we can make scientific progress low-cost and fast

with the help of agent-based models

Outline

Agent-based Modeling Overview

Understanding Performance Bottlenecks

BeSA

Future Steps

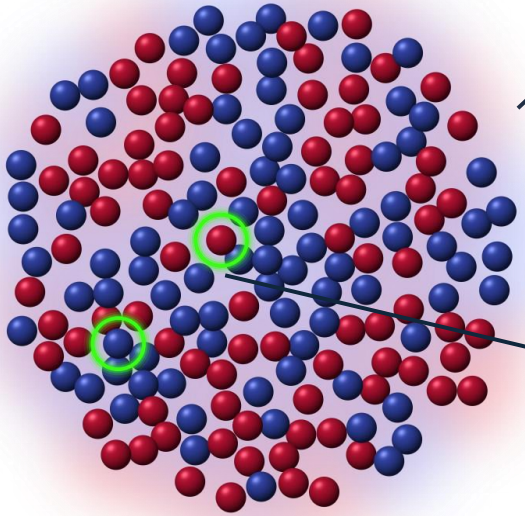
Agent-based Modeling

- A bottom-up simulation technique modeling real-world complex systems
- **Agent**: an autonomous individual with own state and rules
- **Environment:** where agents live
- **Interactions:** agent-agent and agent-environment
- Simulate ABM (agents, environment, interactions)
- **Simulation output:** Large-scale behavioral pattern
- **Simulation step (i.e., iteration):**
  - For each interaction an agent updates its own state
  - Each update is **small**, but is repeated for **millions of agents**
  - Neighbors of an agent **change** as the agent moves

- Simulation = many iterations

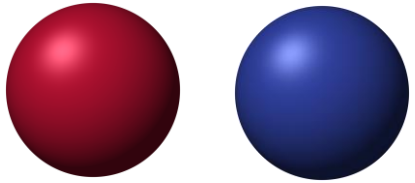
Soma Clustering Example

GOAL: Pull cell bodies tightly together in space to connect and form functional structures



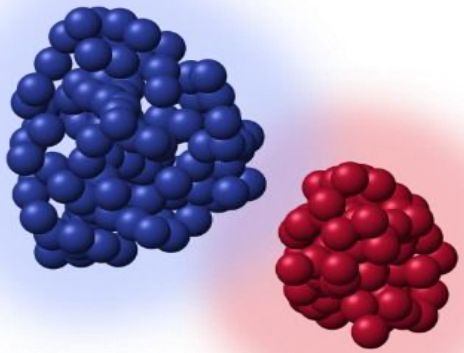
ENVIRONMENT

AGENTS



STATE: Type, Position, Geometry data
RULES: Substance secretion, chemotaxis, mechanical displacement

RESULT



Randomly scattered cells sort themselves into separate and dense colonies over time

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ABM is used in many domains











ABM is used in many domains

Agent-based modeling of erosion and sloughing during growth of *Pseudomonas aeruginosa* biofilms

Check for updates

Ryan Bournes[1] ✉, Suzie Hingley-Wilson[2], Bing Guo[3], Cayla Harris[1], Umar Abubacar[1], Mark Chambers[2] & Roman Bauer[1] ✉

July 2026

June 2026

The EU needs to back its ambition to end animal testing with cash

The European Union has declared that it wants to stop using animals in chemical safety testing. Its goal will need a timeline and a serious funding commitment.

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With great ABMs comes great health research

Large agent count = finer details

Many iterations = looking further into the future

Faster simulations = faster decisions

But, ABMs are not great yet

- Performance bottlenecks are unknown
- Porting to different platforms is not studied (e.g., PiM, GPU)
- Scaling strategies are unclear: scale up or scale out?
- Without this understanding, ABMs stay limited
  - Limited complexity: up to half a trillion agents
  - Limited scalability: requires access to gigantic server clusters

Our Goal

Understand and improve the performance of large-scale agent-based simulations

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BeSA

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BioDynaMo

- The state-of-the-art open-source framework
- Written in C++
- Fully parallel, built for multi-core CPUs
- NUMA-aware
- 1.72 billion agents on a single server
  - One to three orders of magnitude larger than state-of-the-art ABM platforms



BioDynaMo/biodynamo

BioDynaMo is a high-performance and modular, agent-based simulation platform.

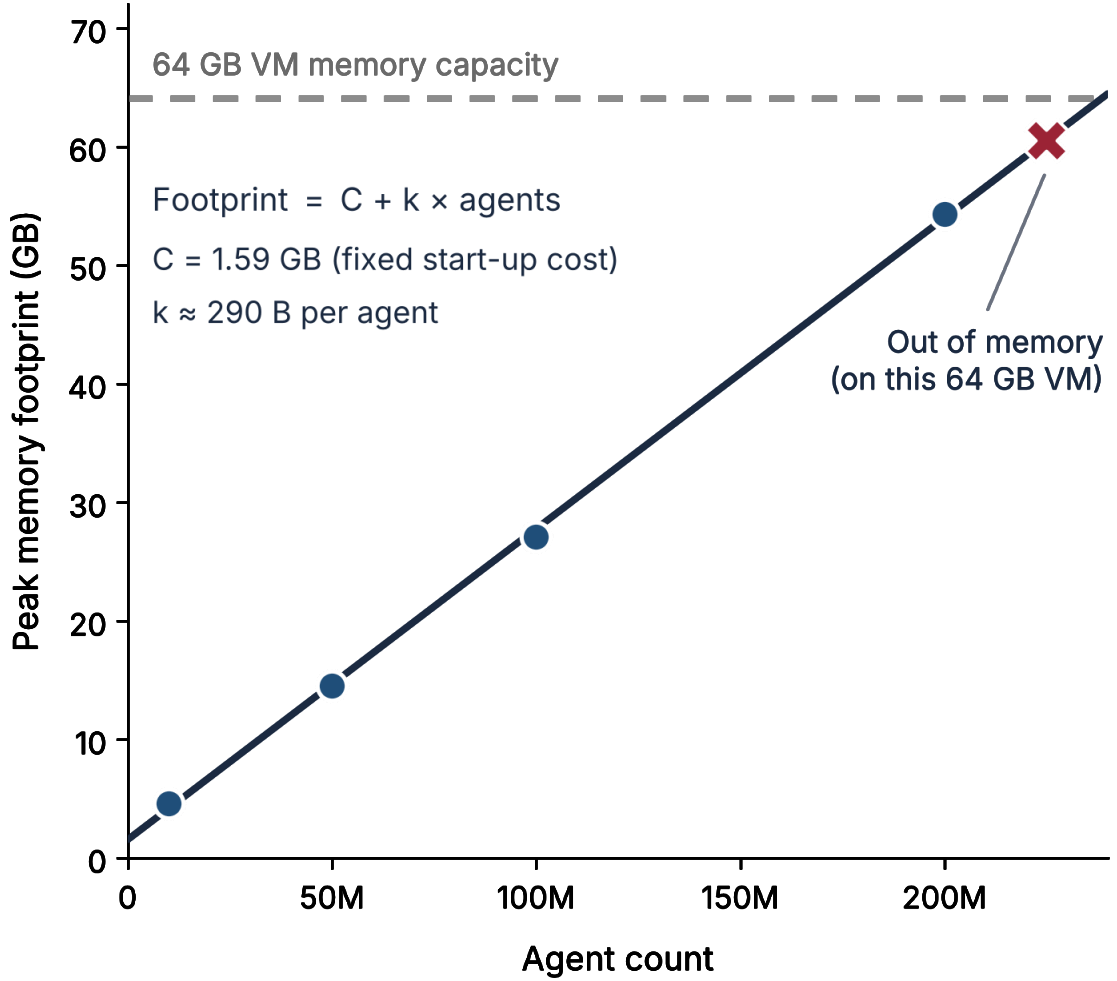
simulation · cancer · biology · high-performance · parallel

C++ · ☆ 137 · Updated 2 days ago

Experimental Setup

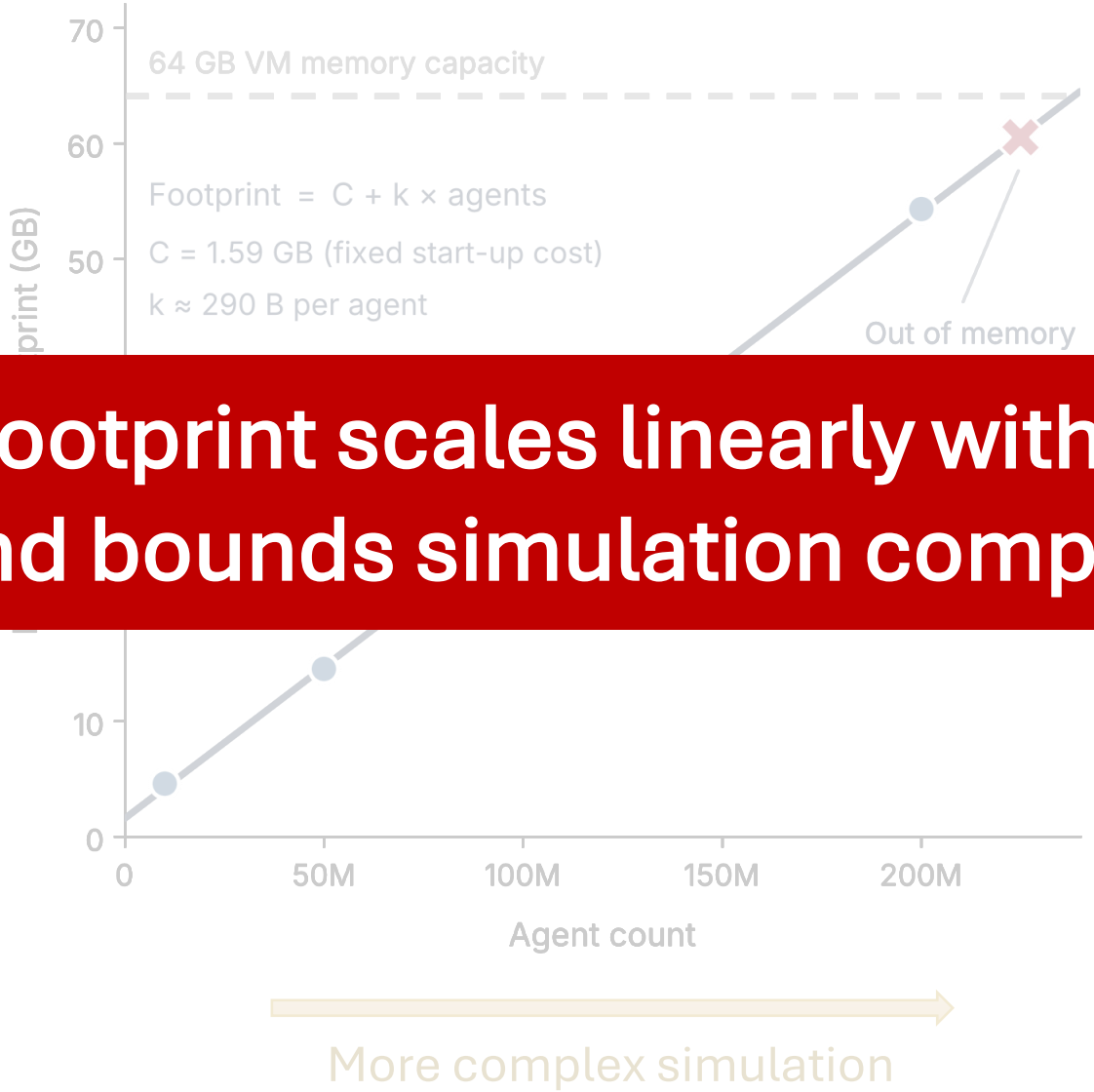
Aspect	Details
Framework	BioDynaMo
Use case	Epidemiology simulation
Scale	10M – 1B agents
Repetitions	3 runs per agent count
Platform	Google Cloud C4 VMs, 16 vCPUs
Methodology	Top-down analysis
Tooling	Perf, sampled at 1 Hz
Derived metrics	IPC, L2 MPKI, memory-bound fraction

Memory Consumption



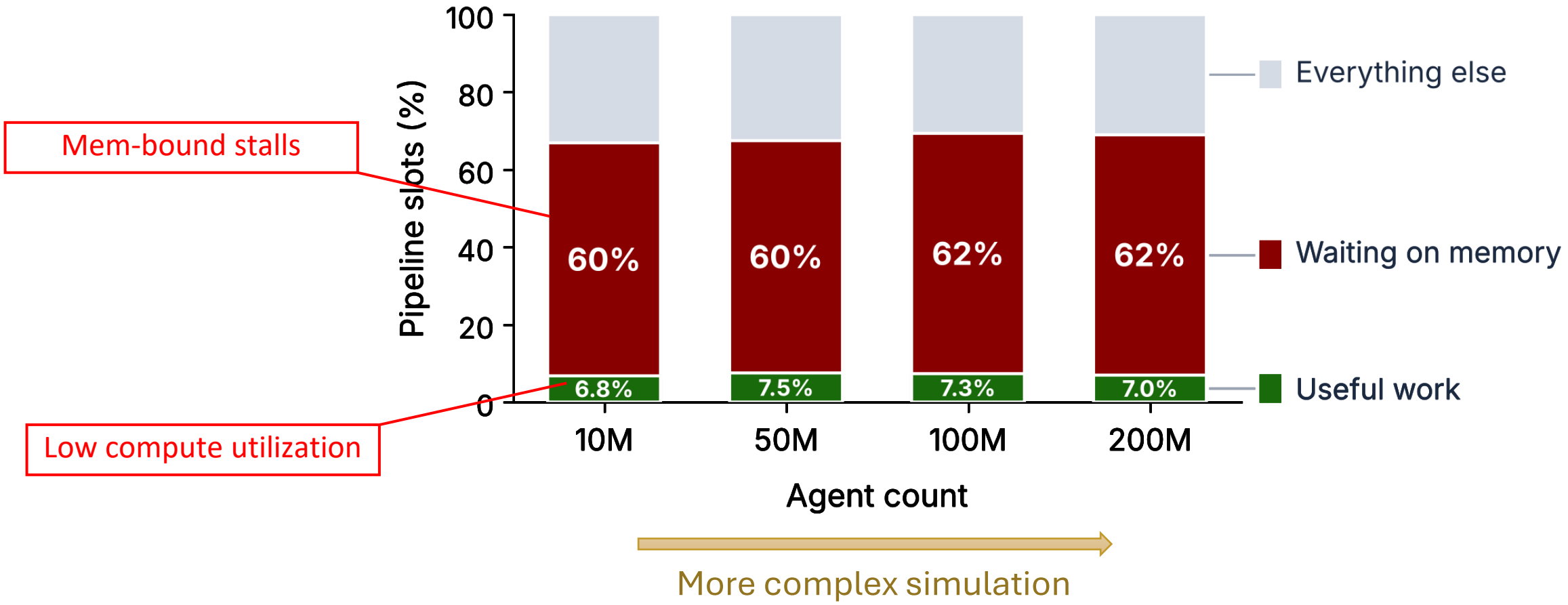
More complex simulation

Memory Consumption

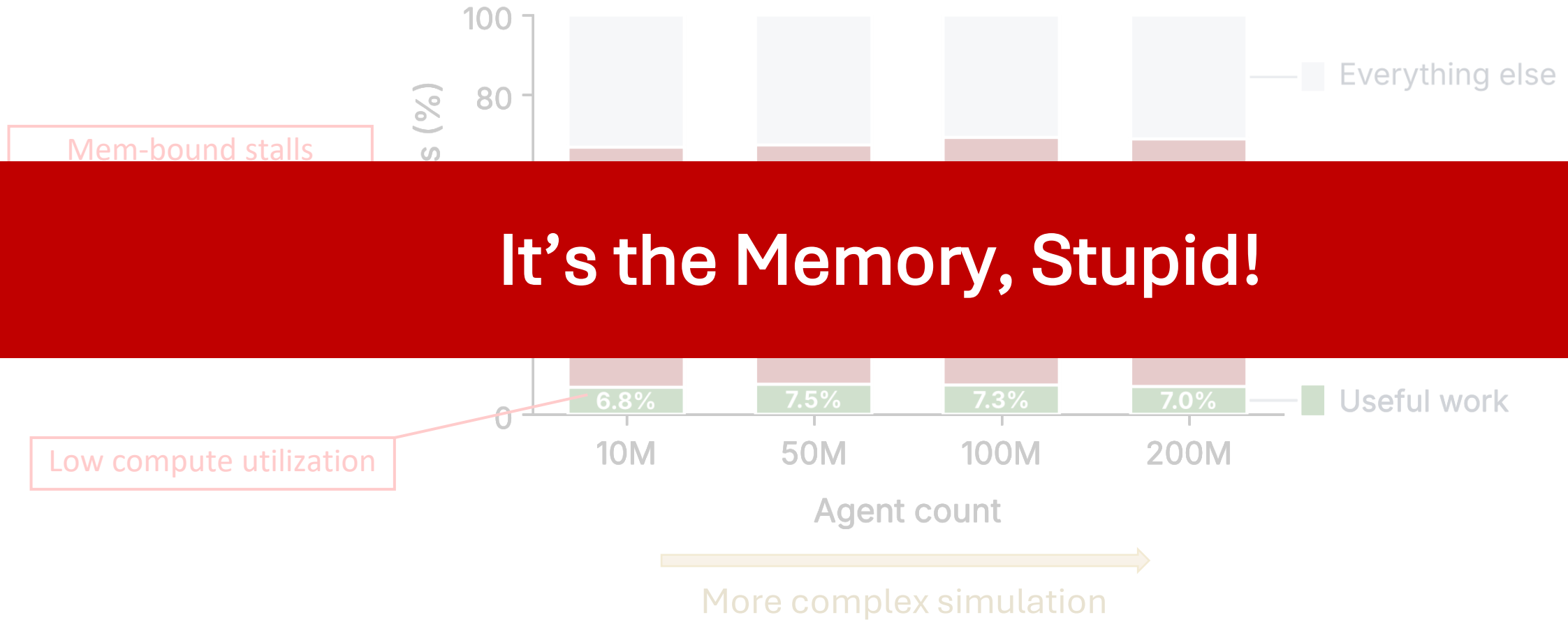


Memory footprint scales linearly with agent count and bounds simulation complexity

Microarchitectural View



Microarchitectural View



It’s the Memory, Stupid!

The Bottleneck

- Low compute utilization + Mem-bound stalls
- ABMs are memory bound
- What type of memory-bound?
  - Bandwidth?
  - Latency?
  - Capacity?
  - Contention?
- Where to optimize?
- Software or Hardware improvements? Both?

We need to know more

Searching for the memory-bound cause

- Roofline analysis
- Full DAMOV methodology
- Few, aggregated metrics on fixed hardware cannot reveal the cause
- Let's use computer architecture simulators
- Simulate the full ABM
  - Far too slow
- Use benchmarks then!



But **no benchmarks** for ABMs anywhere


Houston, we have a problem

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Benchmarks, assemble!

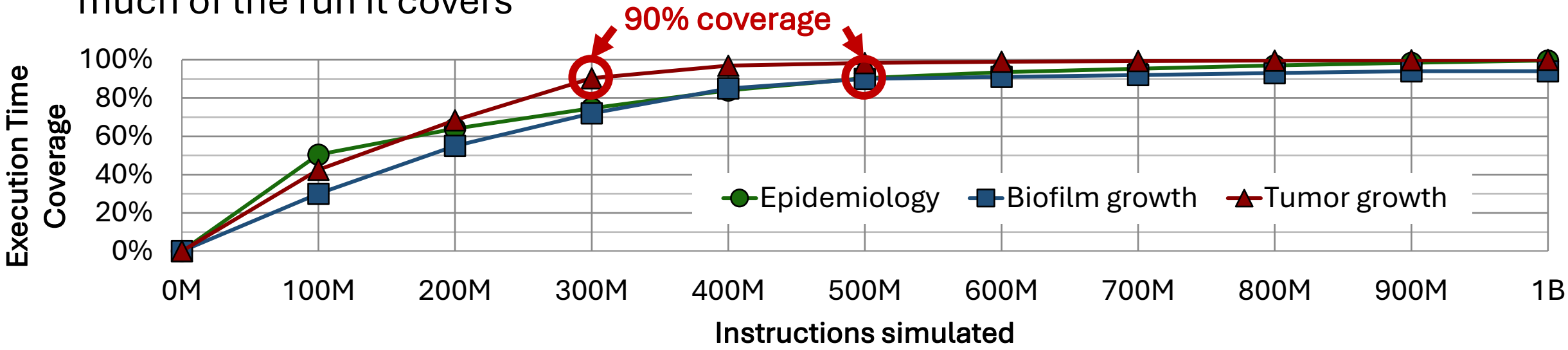
- **BeSA: Be**nchmarks for **S**imulating **A**gent-based Models in architecture research
- Built on BioDynaMo
- Two design goals
  - Representative
  - Reusable
- Fixed agent count
  - The same agents exist from the first iteration to the last
  - Epidemiology
- Dynamic agent count
  - Agents are created or removed as the simulation runs
  - Tumor growth, where agents are cells
  - Biofilm growth, where agents are bacteria



Besa in Albanian means "word of honor", an unbreakable promise that carries deep cultural and moral weight

BeSA

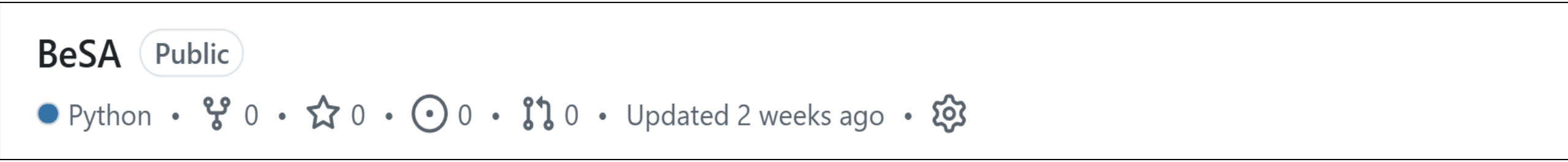
- **Problem:** full ABM runs are too long to simulate cycle-accurately
  - Epidemiology run: ~3.4×10^{14} dynamic instructions
- SimPoint-based sampling
  - **Profile:** basic-block vectors (BBVs) with Valgrind, over intervals of 100M instructions
  - **Cluster:** grouped intervals that execute similar code and weighted each group by how much of the run it covers



A small subset of instructions is representative of the whole execution

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BeSA



https://github.com/SAL-Research/BeSA



https://github.com/SAL-Research/gem5-BeSA

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Future Steps

Future Steps

- Extend BeSA with more use cases
  - Add models from clinical and biomedical research
- Multi-threaded sampling
  - SimPoint assumes a single, reproducible instruction stream
  - Parallel runs require thread alignment
- Native support for trace-based memory simulation (Ramulator 2.1)

A complete benchmark pipeline

Conclusion

- Agent-based modeling (ABM) is a powerful simulation tool used in health research
- **Key Observations:**
  - Short computation wrapped around irregular walks over a neighbor structure
  - Most of the execution time goes to waiting for memory
  - Limited by the memory capacity
- **What type of memory-boundness: bandwidth, latency, capacity or contention?**
- **BeSA:**
  - The first ABM benchmark suite for architecture research (work in progress)
  - Covers different health use cases
  - Compatible with gem5 simulator

We have a lot more to do



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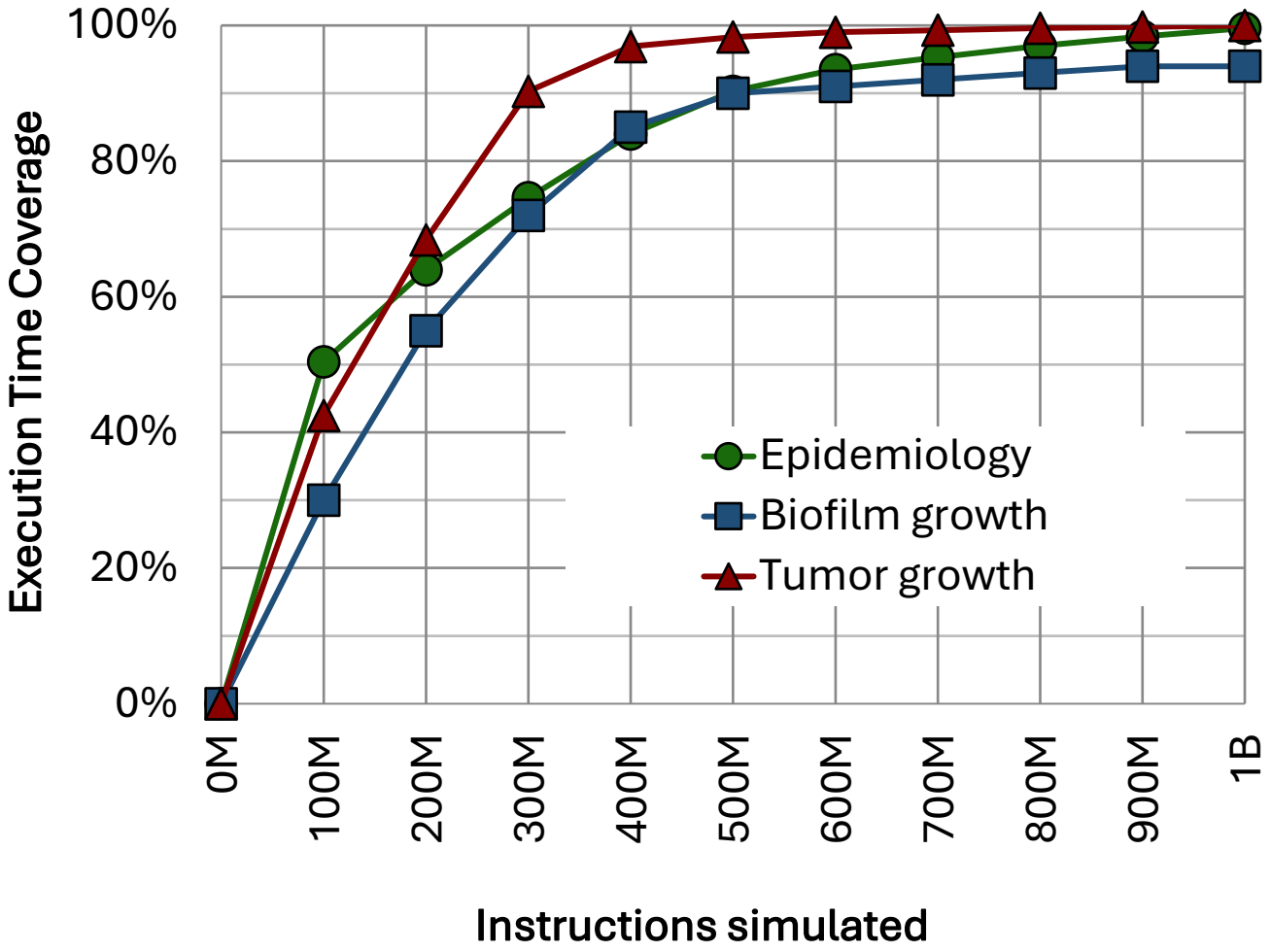
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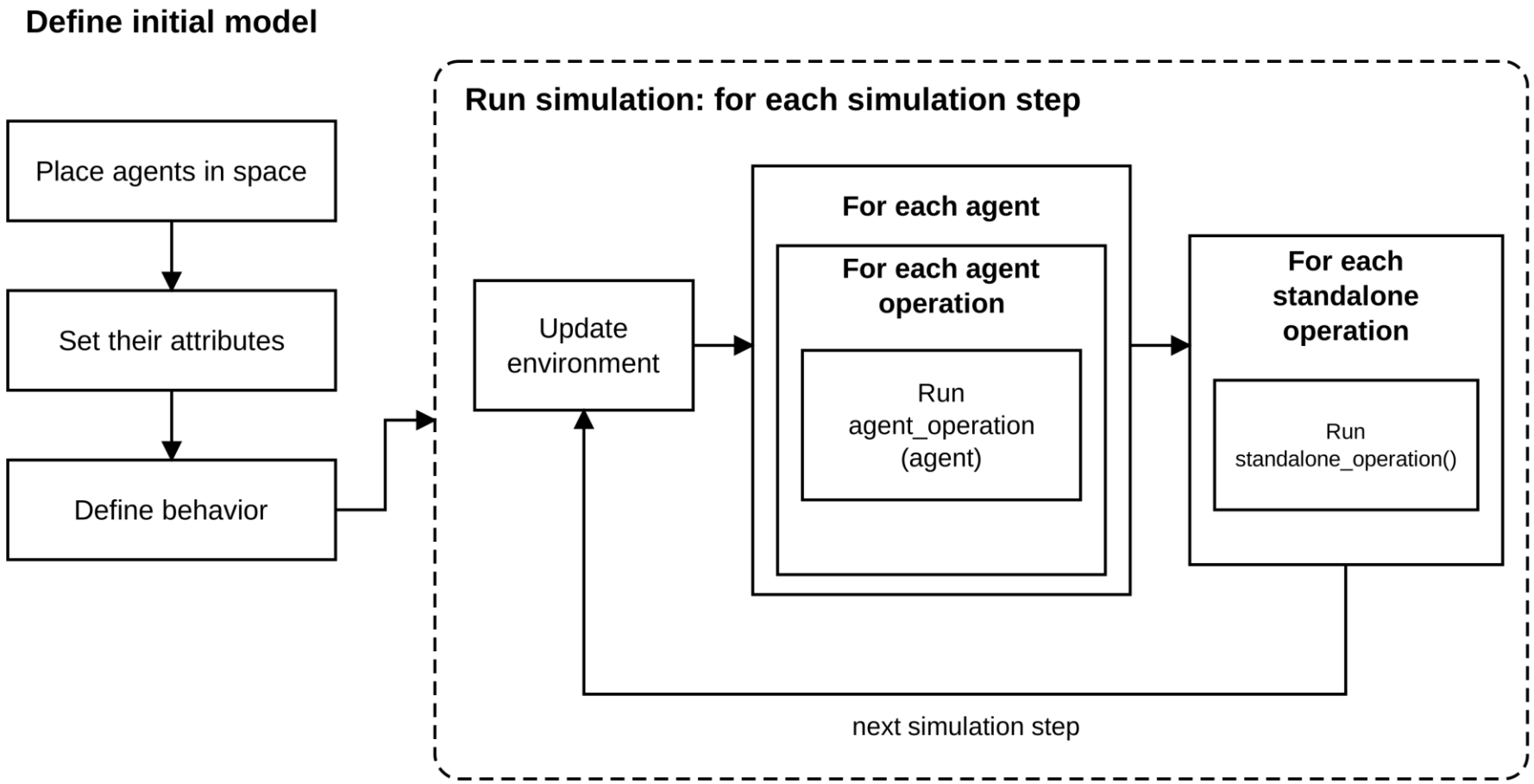
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Execution Time Coverage

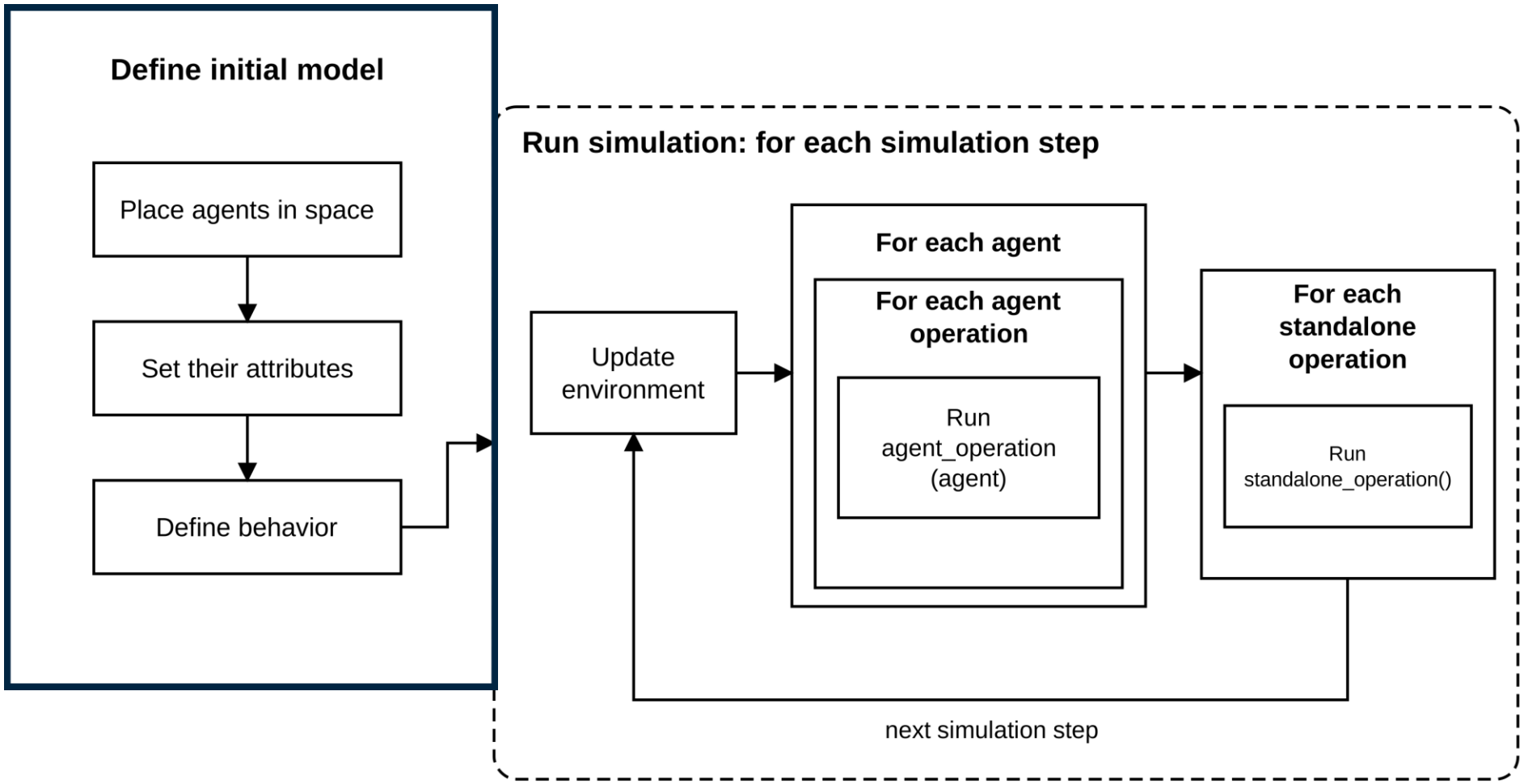
Instr. Count	Epidemiology	Biofilm	Tumor
0	0	0	0
100	0.504	0.3	0.425
200	0.64	0.55	0.684
300	0.746	0.72	0.903
400	0.84	0.85	0.969
500	0.903	0.9	0.983
600	0.935	0.91	0.99
700	0.953	0.92	0.993
800	0.97	0.93	0.996
900	0.984	0.94	0.998
1000	0.996	0.94	1



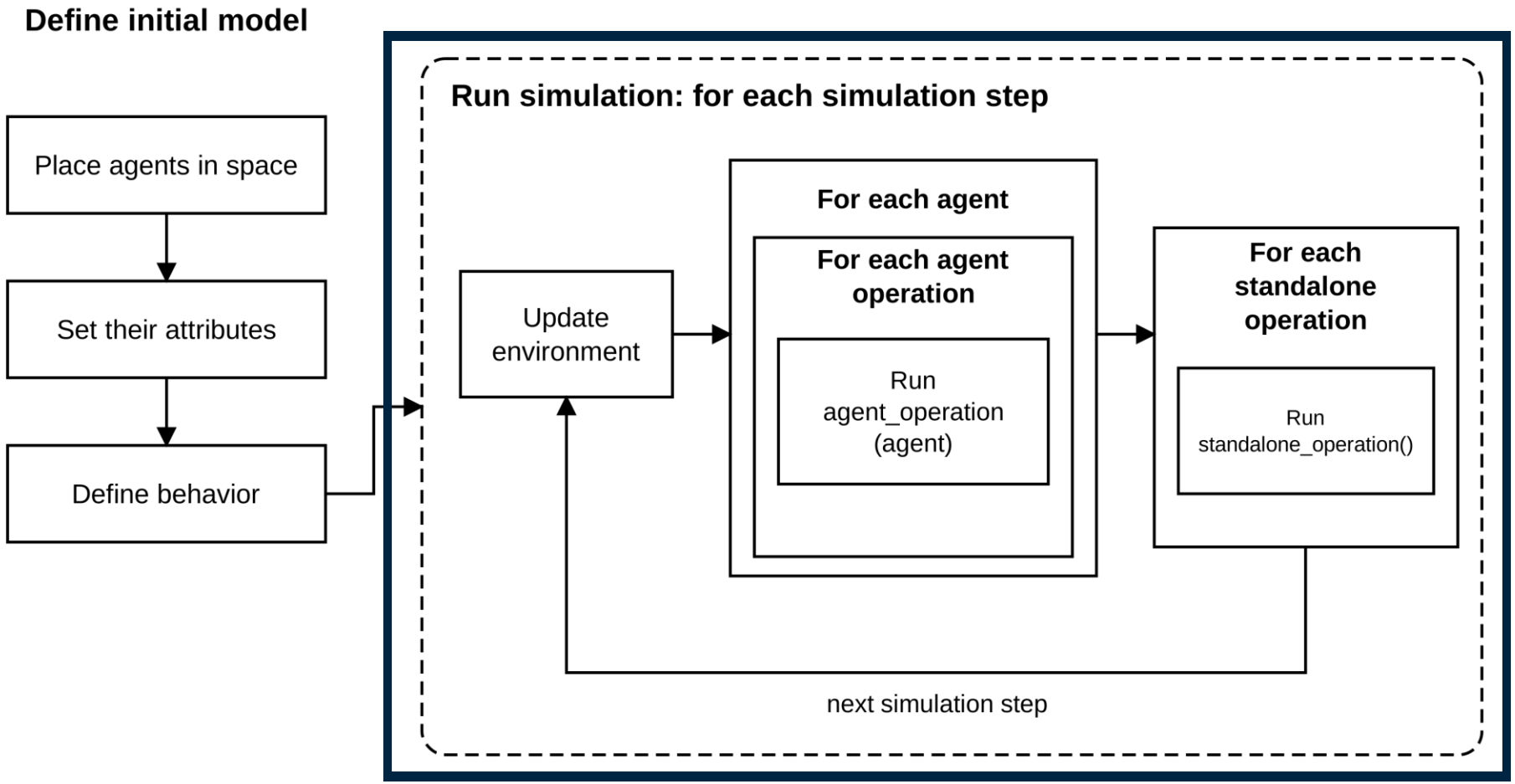
How a simulation runs



How a simulation runs (1)



How a simulation runs (2)



BarrierPoint

- The multi-threaded counterpart of SimPoint
- Uses global barriers in a multi-threaded program to find the most representative inter-barrier regions
- Inter-barrier regions grow with input size
- Estimates total execution time and metrics by simulating only those regions in detail

The pipeline

- Region signatures. Per inter-barrier region, collect basic block vectors (code behavior) and LRU stack-distance vectors (data reuse)
- Clustering. The signatures are normalized, and clustered with instruction-weighted k-means. One representative region per cluster becomes a barrierpoint
- Detailed simulation. Each barrierpoint is simulated independently and in parallel, starting from a checkpoint or a fast-forward to its opening barrier
- Reconstruction. Each barrierpoint gets a multiplier, and metrics are rebuilt as a multiplier-weighted sum

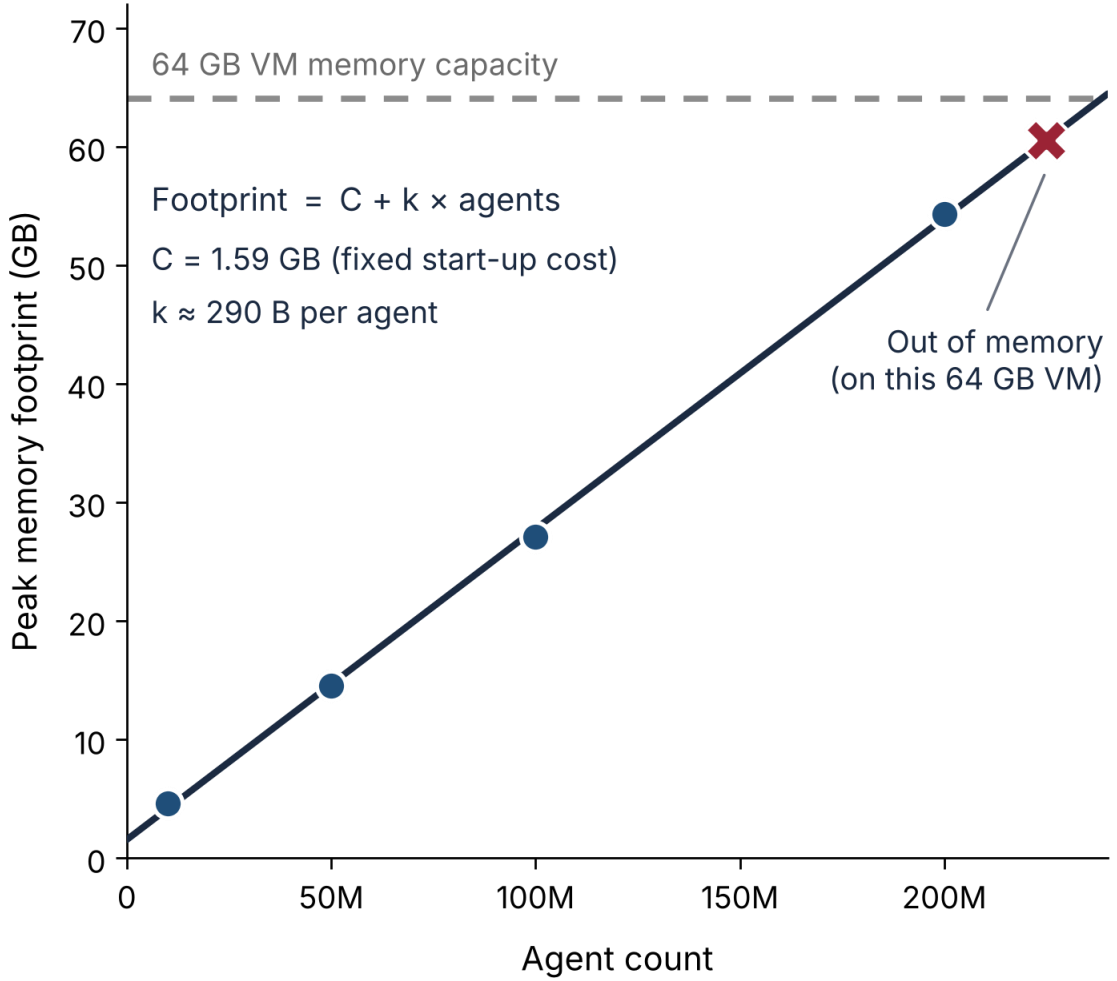
LoopPoint

- BarrierPoint can give almost no speedup when a large application has few barriers
- Uses loop iterations as the unit of work, and regions are bounded by a target size rather than by barrier placement

The pipeline

- Region signatures. The application is recorded once as a PinPlay checkpoint, and a dynamic control-flow graph identifies loops
- Spin-loop filtering. Instructions from the synchronization library (libiomp5 in the paper) are executed but not counted as work
- Slicing. Each slice is roughly N × 100M global instructions for N threads. A region is specified by a (PC, count) pair
- Signatures and clustering. Basic block vectors are collected per thread and concatenated. They are projected to 100 dimensions and clustered with k-means plus a BIC criterion. One representative per cluster becomes a looppoint
- Simulation. Looppoints are simulated independently and in parallel, then runtime is rebuilt as a multiplier-weighted sum over filtered instruction counts

CXL as a solution to the memory consumption limit?



- Why not use CXL?

CXL as a solution to the memory consumption limit?

- A fix for out of memory runs, but not a scaling strategy for scaling to real-world simulations
- We cannot buy our way past the wall
- Latency
- Wrong access pattern (the neighbor search is the hot spot)

Neighbor search

